Marginalization keeps the matching blocks

$$x \sim \mathcal{N}(\mu, \Sigma)$$

$$p(x_1) = \int p(x_1, x_2)\, \mathrm{d}x_2$$

$$= \mathcal{N}(x_1 \mid \mu_1, \Sigma_{11})$$

$$p(x_2) = \mathcal{N}(x_2 \mid \mu_2, \Sigma_{22})$$

Marginal proof

Select x_1: $P = [I \;\; 0]$, $\quad x_1 = Px$

By the affine rule:

$$x_1 \sim \mathcal{N}(P\mu, P\Sigma P^T)$$

Murphy, Book 2 §2.3.1.3, Eqs. 2.76–2.77; affine rule: Book 1 §3.1.1, Eq. 3.5

- Marginalizing x_2 integrates over all its possible values
- Cross-covariance affects conditioning, not the marginal of x_1

Conditioning shifts the mean by an observed deviation

Centered residual

$$B = \Sigma_{12}\Sigma_{22}^{-1}, \quad r = x_1 - \mu_1 - B(x_2 - \mu_2)$$
$$E[r] = 0$$
$$\mathrm{Cov}[r, x_2] = \Sigma_{12} - B\Sigma_{22} = 0$$

Affine closure: (r, x_2) is jointly Gaussian

Zero cross-covariance gives r independent of x_2

$$E[r \mid x_2] = E[r] = 0$$

$$E[x_1 \mid x_2] = \mu_1 + B(x_2 - \mu_2) = \mu_{1|2}$$

Murphy, Book 2 §2.3.1.3, Eq. 2.78; course residual proof uses affine Gaussian closure

- The observed deviation $x_2 - \mu_2$ shifts the mean through $B = \Sigma_{12}\Sigma_{22}^{-1}$.

Conditioning removes explained covariance

$$\Sigma_{1|2} = \Sigma_{11} - \Sigma_{12}\Sigma_{22}^{-1}\Sigma_{21}$$

$$\Sigma_{2|1} = \Sigma_{22} - \Sigma_{21}\Sigma_{11}^{-1}\Sigma_{12}$$

Covariance proof

$$B = \Sigma_{12}\Sigma_{22}^{-1}$$

$$r = x_1 - \mu_{1|2}$$

From the mean proof: r and x_2 are independent

Murphy, Book 2 §2.3.1.3, Eqs. 2.78–2.79; proof reuses the independent residual

- Conditioning cannot increase variance in any direction
- In a Gaussian, this covariance depends on which variables are observed, not their values